

ADVANCED QUANTUM MECHANICS

Practice exam 16.02.2026



TECHNISCHE
UNIVERSITÄT
DARMSTADT

Part 1: true/false questions

Each statement is either **True** or **False**. For each statement you get **0.2 points** if your choice is correct and **0 points** otherwise (no negative points). There are 20 questions with 5 statements each, so the maximum score is

$$0.2 \times 5 \times 20 = 20 \text{ Points.}$$

EXAMPLE: Uncertainty in quantum mechanics

1. The uncertainty relation is a statement about measurement errors of the apparatus. ☐ True ☒ False
2. The uncertainty relation can be violated if measurements are performed fast enough. ☐ True ☒ False
3. Commuting observables necessarily have a nonzero uncertainty product. ☒ True ☐ False
4. A Gaussian wavepacket can saturate the Heisenberg bound. ☒ True ☐ False
5. $\Delta x \Delta p = 0$ is possible for a normalizable state. ☐ True ☒ False

Score: 4/5 correct $\Rightarrow$ 0.8 points.

Fundamental quantum mechanics

F1. Operators and observables

1. Every Hermitian operator has only positive eigenvalues. ☐ True ☐ False
2. Two non-commuting operators can always be simultaneously diagonalized by a suitable basis choice. ☐ True ☐ False
3. The expectation value of an observable is always equal to one of its eigenvalues. ☐ True ☐ False
4. If a system is in an eigenstate of an operator at $t = 0$, it will remain in that eigenstate for any Hamiltonian. ☐ True ☐ False
5. If $[\hat{A}, \hat{H}] = 0$, then the expectation value $\langle \hat{A} \rangle$ is constant in time for any state. ☐ True ☐ False

F2. Quantum harmonic oscillator

1. The ground-state energy of the harmonic oscillator is $\frac{1}{2}\hbar\omega$. ☐ True ☐ False
2. The spacing between consecutive energy levels is constant. ☐ True ☐ False
3. The expectation value $\langle x \rangle$ in the ground state is nonzero. ☐ True ☐ False
4. Ladder operators a and $a^\dagger$ are Hermitian operators. ☐ True ☐ False
5. The ground-state wavefunction is a Gaussian in position space. ☐ True ☐ False

F3. Hydrogen atom

1. The energy levels depend only on the principal quantum number n . ☐ True ☐ False
2. States with different ℓ but same n are degenerate in the Coulomb potential. ☐ True ☐ False
3. The ground state has orbital angular momentum $\ell = 1$. ☐ True ☐ False
4. In a central potential, the Schrödinger equation does not separate into radial and angular parts. ☐ True ☐ False
5. The degeneracy of a level with quantum number n (ignoring spin) is n^2 . ☐ True ☐ False

F4. Time-independent perturbation theory

1. First-order energy corrections are expectation values of the perturbation in the unperturbed eigenstates. ☐ True ☐ False
2. Degenerate perturbation theory requires diagonalizing the perturbation in the degenerate subspace. ☐ True ☐ False
3. First-order energy shifts always vanish if the perturbation is odd and the state has definite parity. ☐ True ☐ False
4. The perturbed eigenstates remain orthogonal to all unperturbed eigenstates. ☐ True ☐ False
5. Perturbation theory gives exact results for arbitrarily large perturbations. ☐ True ☐ False

F5. Particle in a one-dimensional box

1. The energy spectrum is discrete. ☐ True ☐ False
2. The ground-state energy is zero. ☐ True ☐ False
3. The wavefunction vanishes at the walls of the box. ☐ True ☐ False
4. Energy levels scale as n^2 . ☐ True ☐ False
5. The probability density in the first excited state is uniform in space. ☐ True ☐ False

Scattering theory

S1. Born approximation

1. The Born approximation is valid for weak potentials. ☐ True ☐ False
2. In the Born approximation, multiple scattering processes are neglected. ☐ True ☐ False
3. The Born approximation always gives exact results. ☐ True ☐ False
4. The scattering amplitude in first Born order involves an integral over the potential. ☐ True ☐ False
5. The Born approximation requires a spherically symmetric potential. ☐ True ☐ False

S2. Momentum transfer

1. The momentum transfer is defined as $\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i$. ☐ True ☐ False
2. Elastic scattering implies $|\mathbf{k}_f| = |\mathbf{k}_i|$. ☐ True ☐ False
3. The scattering angle is independent of the momentum transfer. ☐ True ☐ False
4. For elastic scattering $q = 2k \sin(\theta/2)$. ☐ True ☐ False
5. Momentum transfer has dimensions of energy. ☐ True ☐ False

S3. Partial waves

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|---|-------------------------------|--------------------------------|
| 1. A plane wave can be decomposed into spherical harmonics. | ☐ True | ☐ False |
| 2. Each angular momentum channel scatters independently. | ☐ True | ☐ False |
| 3. Phase shifts contain information about the potential. | ☐ True | ☐ False |
| 4. Only the $\ell = 0$ partial wave exists at all energies. | ☐ True | ☐ False |
| 5. At low energies, the s-wave dominates scattering. | ☐ True | ☐ False |

S4. Cross sections

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| 1. The total cross section has units of area. | ☐ True | ☐ False |
| 2. The differential cross section depends on angle. | ☐ True | ☐ False |
| 3. The cross section measures the physical size of the scatterer exactly. | ☐ True | ☐ False |
| 4. Integrating the differential cross section gives the total cross section. | ☐ True | ☐ False |
| 5. Cross sections can be measured experimentally. | ☐ True | ☐ False |

S5. General scattering properties

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| 1. Scattering requires a nonzero interaction potential. | ☐ True | ☐ False |
| 2. The incoming state is typically taken as a plane wave. | ☐ True | ☐ False |
| 3. The scattering amplitude contains angular information. | ☐ True | ☐ False |
| 4. The scattered wave is localized near the origin at large distances. | ☐ True | ☐ False |
| 5. Scattering theory applies only in one spatial dimension. | ☐ True | ☐ False |

Many body physics

M1. Indistinguishable particles

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| 1. For indistinguishable particles the probability density is invariant under particle exchange. | ☐ True | ☐ False |
| 2. Fermionic wavefunctions are symmetric under exchange of particle labels. | ☐ True | ☐ False |
| 3. Bosons obey the Pauli exclusion principle. | ☐ True | ☐ False |
| 4. Identical fermions are allowed to occupy exactly the same quantum state. | ☐ True | ☐ False |
| 5. The exchange of two particles always produces an observable change in all measurements. | ☐ True | ☐ False |

M2. Bosons and fermions

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| 1. Bosonic many-body states are symmetric combinations of single-particle states. | ☐ True | ☐ False |
| 2. A Slater determinant describes a system of bosons. | ☐ True | ☐ False |
| 3. Fermionic states are antisymmetric under particle exchange. | ☐ True | ☐ False |
| 4. Two fermions in the same spin state can occupy the same orbital. | ☐ True | ☐ False |
| 5. Bosons can macroscopically occupy a single quantum state. | ☐ True | ☐ False |

M3. Field operators

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| 1. Field operators create or destroy particles at a position in space. | ☐ True | ☐ False |
| 2. Field operators can be expanded in a basis of single-particle modes. | ☐ True | ☐ False |
| 3. Field operators exist only for bosons. | ☐ True | ☐ False |
| 4. The Hamiltonian can be written in terms of field operators. | ☐ True | ☐ False |
| 5. Field operators automatically enforce the correct particle statistics. | ☐ True | ☐ False |

M4. Helium atom

1. The helium atom contains electron–electron interaction. ☐ True ☐ False
2. The Hamiltonian separates exactly into two independent hydrogen atoms. ☐ True ☐ False
3. The ground state must respect fermionic antisymmetry. ☐ True ☐ False
4. A variational parameter can mimic screening of the nucleus. ☐ True ☐ False
5. The helium atom has no spin degrees of freedom. ☐ True ☐ False

M5. Lattice spin models (Heisenberg and Ising)

1. The Heisenberg model describes interacting quantum spins on a lattice. ☐ True ☐ False
2. The transverse-field Ising model introduces quantum fluctuations via a field proportional to σ^x . ☐ True ☐ False
3. Exchange coupling in the Heisenberg model comes from classical magnetic dipole forces only. ☐ True ☐ False
4. The Ising Hamiltonian contains only kinetic-energy terms and no spin interactions. ☐ True ☐ False
5. The antiferromagnetic Heisenberg model has a triplet ground state for two spins. ☐ True ☐ False

Relativistic quantum mechanics

R1. Lorentz invariance

1. The relativistic dispersion relation is $E^2 = p^2 c^2 + m^2 c^4$. ☐ True ☐ False
2. The spacetime interval is invariant under Lorentz transformations. ☐ True ☐ False
3. Galilean transformations preserve the speed of light. ☐ True ☐ False
4. In the non-relativistic limit $|p| \ll mc$, the energy approaches $mc^2 + \frac{p^2}{2m}$. ☐ True ☐ False
5. Time and space coordinates are independent absolute quantities in special relativity. ☐ True ☐ False

R2. Dirac equation and Dirac field

1. The Dirac equation is linear in both time and spatial derivatives. ☐ True ☐ False
2. The adjoint spinor is defined as $\bar{\psi} = \psi^\dagger \gamma^0$. ☐ True ☐ False
3. The Dirac Lagrangian contains no mass term. ☐ True ☐ False
4. The Dirac equation describes only scalar particles. ☐ True ☐ False
5. Quantizing the Dirac field produces only electron states and no antiparticles. ☐ True ☐ False

R3. Antiparticles

1. Negative energy states can be interpreted as antiparticles. ☐ True ☐ False
2. Antiparticles have the same mass but opposite charge. ☐ True ☐ False
3. The Dirac equation predicts the existence of the positron. ☐ True ☐ False
4. Antiparticles appear only in non-relativistic quantum mechanics. ☐ True ☐ False
5. The relativistic spectrum contains both positive and negative energy branches. ☐ True ☐ False

R4. Klein paradox

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| 1. The reflection coefficient can exceed unity in relativistic scattering. | ☐ True ☐ False |
| 2. The paradox indicates pair creation processes. | ☐ True ☐ False |
| 3. The paradox can be explained using only non-relativistic quantum mechanics. | ☐ True ☐ False |
| 4. The Klein region occurs only for weak potentials. | ☐ True ☐ False |
| 5. The paradox suggests particle number conservation fails. | ☐ True ☐ False |

R5. Quantum electrodynamics

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| 1. QED describes interactions between charged fermions and photons. | ☐ True ☐ False |
| 2. The electromagnetic field in QED is represented by a scalar potential only. | ☐ True ☐ False |
| 3. Photon emission and absorption violate energy conservation. | ☐ True ☐ False |
| 4. Pair creation and annihilation are forbidden in QED. | ☐ True ☐ False |
| 5. Electrons are classical point particles and not quantum excitations of a field. | ☐ True ☐ False |

Part 2: Exercises

Each exercise is **5 points**. Maximum score for part 2 is

$$5 \times 2 = 10 \text{ Points.}$$

Exercise 1: Two-site Bose–Hubbard model

Consider bosons in a two-site lattice (modes 1, 2) with Hamiltonian

$$\hat{H} = -J(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1) + \frac{U}{2} \sum_{j=1}^2 \hat{n}_j(\hat{n}_j - 1), \quad \hat{n}_j = \hat{a}_j^\dagger \hat{a}_j,$$

in the fixed particle-number sector $N = 2$.

We use the Fock basis

$$|2, 0\rangle, \quad |1, 1\rangle, \quad |0, 2\rangle.$$

(i) Matrix representation

Using

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle, \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle,$$

compute the action of each term of $\hat{H}$ on the three basis states and construct the 3×3 matrix representation of $\hat{H}$.

(ii) Non-interacting limit $U = 0$

Diagonalize the Hamiltonian for $U = 0$ (use symmetry under exchange of the two sites).

Identify:

- the ground state,
- the highest-energy state.

(iii) Interaction-dominated limit $J = 0$

For $J = 0$, the Hamiltonian becomes diagonal in the Fock basis.

- (i) Compute the energy of each basis state.
- (ii) Determine the ground state for:

- $U > 0$ (repulsive interaction),
- $U < 0$ (attractive interaction).

Explain physically why the bosons either separate or bunch together.

(iv) **Symmetry reduction**

Notice that the Hamiltonian is invariant under exchange of the two sites.

Define symmetric and antisymmetric states

$$|S\rangle = \frac{|2, 0\rangle + |0, 2\rangle}{\sqrt{2}}, \quad |A\rangle = \frac{|2, 0\rangle - |0, 2\rangle}{\sqrt{2}}.$$

Show that $|A\rangle$ is an eigenstate of $\hat{H}$ and find its energy.

Reduce the problem to a 2×2 Hamiltonian in the subspace $\{|S\rangle, |1, 1\rangle\}$.

Exercise 2: Dirac fermion in a 1D infinite-wall box (t, x notation)

The Dirac two-component spinor

$$\psi(t, x) = \begin{pmatrix} \psi_1(t, x) \\ \psi_2(t, x) \end{pmatrix},$$

satisfies the coupled Dirac equations (free particle):

$$i\hbar\partial_t\psi_1 = -i\hbar c\partial_x\psi_2 + mc^2\psi_1, \quad i\hbar\partial_t\psi_2 = -i\hbar c\partial_x\psi_1 - mc^2\psi_2.$$

The particle is confined to an *infinite wall box*

$$0 < x < L, \quad \text{and no probability current may flow through } x = 0, L.$$

We look for stationary states

$$\psi(t, x) = e^{-\frac{i}{\hbar}Et} \phi(x), \quad \phi(x) = \begin{pmatrix} \phi_1(x) \\ \phi_2(x) \end{pmatrix}.$$

(i) **Stationary equations**

Write the stationary Dirac equations satisfied by $\phi_1(x)$ and $\phi_2(x)$.

(ii) **Plane-wave solutions inside the box**

Assume $|E| > mc^2$ and define the (real) momentum

$$p := \frac{1}{c}\sqrt{E^2 - m^2c^4}.$$

Find the equation for ϕ_1 and show that the solution is

$$\phi_1(x) = A e^{ipx/\hbar} + B e^{-ipx/\hbar}.$$

Use the stationary equations to express $\phi_2(x)$ in terms of $\phi_1(x)$.

(iii) **Energy spectrum and non-relativistic limit**

With the quantization of momentum

$$pL = n\pi\hbar, \quad n = 1, 2, 3, \dots$$

the allowed energies are

$$E_n = \pm \sqrt{m^2c^4 + \frac{n^2\pi^2\hbar^2c^2}{L^2}}, \quad n = 1, 2, 3, \dots$$

Consider the positive-energy branch and find the limit of the non-relativistic particle-in-a-box spectrum.